A Lightweight Trust Management Framework in the Networking Layer of IoT Devices.

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Preliminary Progress Review (PPR) Report of 25_26J_009

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Declaration

I declare that this is my own work, and this proposal does not incorporate without acknowledgment any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgment is made in the text.

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1.INTRODUCTION

1.1 Purpose

The purpose of this Preliminary Progress Review (PPR) document is to present the current progress of the research project, including background study, problem identification, research objectives, methodology, and the implementation work completed up to this stage. This document is intended for academic supervisors and evaluators to review the feasibility, direction, and progress of the research

1.2 Definitions, Acronyms, and Abbreviations

Acronym	Description
IoT	Internet of Things
RPL	Routing Protocol for Low-Power and Lossy
LLN	Low-Power and Lossy Network
ETX	Expected Transmission Count
HO	Historical Observation
ER	Residual Energy
DAG	Directed Acyclic Graph
TCE	Trust Computation Engine

1.3 Overview

This document is organized as follows. Section 2 presents the literature review and background study related to trust-based routing in IoT networks. Section 3 identifies the research problem and its significance. Section 4 outlines the research objectives and methodology. Section 5 describes the scope, expected outcomes, and constraints. Section 6 presents the project plan and schedule.

2.LITERATURE REVIEW

IoT networks commonly operate under severe constraints such as limited energy, unreliable wireless links, and dynamic topology changes. RPL is widely adopted as a routing protocol for LLNs; however, it lacks inherent trust and security mechanisms. Recent research has proposed trust-based routing approaches that incorporate metrics such as packet forwarding behavior, energy awareness, and node reliability. While these approaches improve security, many of them introduce high computational overhead or do not consider energy-adaptive trust evaluation. Therefore, there is a need for a lightweight and energy-aware trust management framework suitable for constrained IoT environments.

Identification and significance of the problem

Existing trust-based routing solutions for IoT networks often fail to balance security and energy efficiency. Many approaches rely on frequent trust computation, which significantly drains node energy and reduces network lifetime. Additionally, trust evaluation is commonly static and does not adapt to changing node conditions. The proposed research addresses this gap by introducing an energy-aware and adaptive trust computation framework integrated with RPL, making it suitable for real-world IoT deployments.

3.RESEARCH QUESTIONS

How can trust be evaluated and managed in RPL-based IoT networks in a lightweight and energy-efficient manner while maintaining reliable routing decisions?

4. RESEARCH OBJECTIVES

- Studying existing trust-based routing mechanisms in IoT networks
- To design an energy-aware trust computation framework for RPL
- To implement trust metrics such as Historical Observation and Residual Energy
- To validate the proposed framework using Contiki-NG and Cooja simulation

5. RESEARCH METHODOLOGY

The research follows a simulation-based experimental methodology. Contiki-NG is used as the IoT operating system, and Cooja simulator is employed to evaluate the proposed framework. Initially, a baseline RPL network is implemented using UDP client-server communication. Trust metrics are then implemented incrementally. Historical Observation is derived from ETX values obtained from the RPL routing layer, while Residual Energy is computed using Contiki's ENERGEST module. Trust values are normalized and logged during runtime to evaluate node behavior dynamically.

6. ANTICIPATED BENEFITS

6.1 Benefits to Users

The proposed trust management framework provides several benefits to users of IoT networks. By integrating trust evaluation into the RPL routing protocol, the system improves routing reliability by avoiding nodes that exhibit poor communication behavior or low residual energy. The use of lightweight trust metrics ensures that trust computation does not impose significant computational or energy overhead on resource-constrained IoT devices. Additionally, the energy-aware design helps extend network lifetime by reducing unnecessary trust computations and preventing premature battery depletion of sensor nodes. Overall, the framework enhances network stability, efficiency, and reliability in low-power IoT environments.

6.2 Contribution to Knowledge

This research contributes to the existing body of knowledge by proposing an energy-aware trust computation framework specifically designed for RPL-based IoT networks. Unlike traditional trust models that rely on frequent and static trust evaluation, the proposed approach integrates Historical Observation derived from real-time routing metrics and Residual Energy calculated using Contiki's ENERGEST module. The implementation demonstrates how trust can be computed dynamically during runtime without modifying the core RPL protocol. Furthermore, the research provides practical insights into implementing trust metrics within Contiki-NG and validating them through Cooja simulation, offering a reproducible and extensible foundation for future trust-based IoT research.

7. SCOPE AND EXPECTED OUTCOMES

The research focuses on trust management at the routing layer of IoT networks using RPL. The system evaluates trust based on communication behavior and residual energy. Advanced trust aggregation and attack mitigation mechanisms are considered as future extensions.

8. RESEARCH CONSTRAINTS

- Simulation-based evaluation only
- Limited attack models at early stages
- Energy values based on Contiki energy model

9. PROJECT PLAN OR SCHEDULE

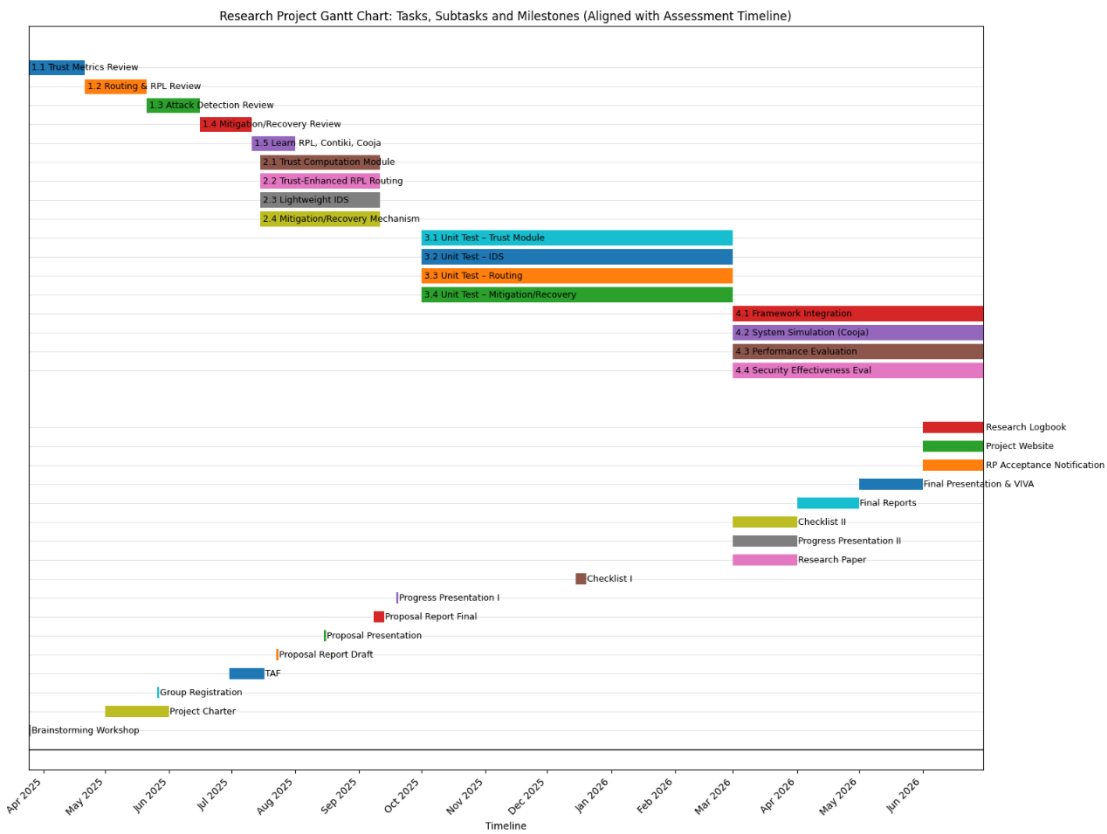

Figure 1: Task and Time Sequence of the Energy-Efficient Trust Computation Engine

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